Quasi-likelihoods

Statistics III - CdL SSE

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Homepage



Meat ants (*Iridomyrmex purpureus*).

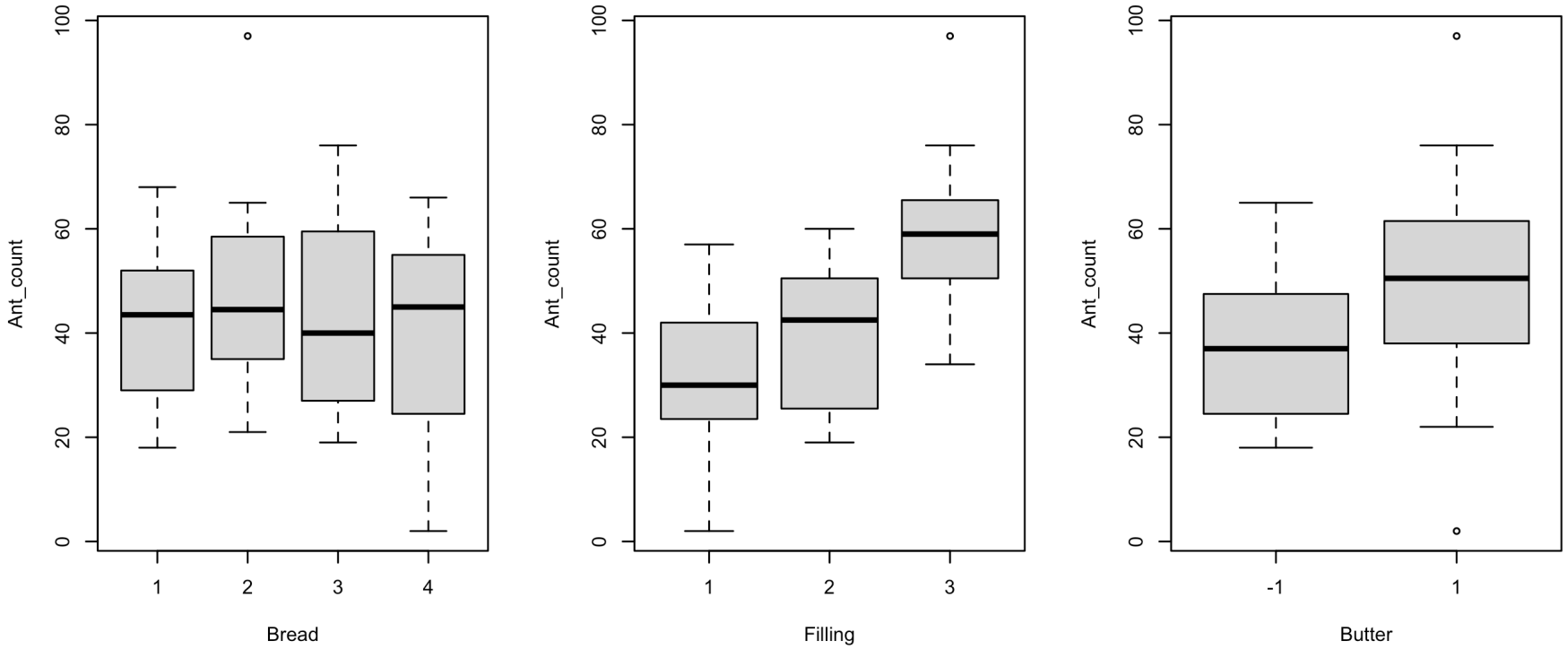
- In Unit C and Unit D we saw that binomial and Poisson regression may suffer from **overdispersion**: the data are more variable than the model allows.
- This unit presents **quasi-likelihoods** (Wedderburn 1974), a simple way to handle overdispersion **without** specifying a distribution for Y_i: we only specify its **mean** and its **variance**.
- The unit has three parts: a **motivating example**, the **theory** of quasi-likelihoods, and the **complete analysis** of the example.
- An alternative, **not covered** in this course, is to replace the Poisson or the binomial with a different parametric distribution, e.g. the negative binomial.

> The content of this Unit is covered in **Chapter 6** of Salvan et al. (2020). Alternatively, see **Sections 8.1 and 8.3** of Agresti (2015).

The `Ants` data

- The `Ants` dataset of the `MLGdata` library comes from an experiment run by students of an Australian university (Salvan et al. 2020). The aim is to study which sandwiches **meat ants** prefer.
- The **explanatory variables** are three experimental factors:
  - `Bread` (4 types): rye, wholemeal, multigrain, white;
  - `Filling` (3 types): Vegemite, peanut butter, ham and pickles;
  - `Butter` (2 levels): present (1) or absent (-1).
- For each of the $4 \times 3 \times 2 = 24$ combinations, **two** pieces of sandwich were placed near the entrance of an ant nest, for a total of $n = 48$ observations.
- The **response** `Ant_count` is the number of ants on the sandwich after 5 minutes.

Exploratory analysis



- `Filling` and `Butter` seem to matter, whereas `Bread` has a weaker effect.
- The response is a **count**, therefore a natural starting point is a **Poisson regression** model with logarithmic link, including the `Filling`-`Butter` interaction ($p = 9$).

Home page

A Poisson regression model

	Estimate	Std. Error	p-value
(Intercept)	3.386	0.075	<0.001
Bread2	0.154	0.061	0.011
Bread3	0.026	0.063	0.684
Bread4	-0.030	0.064	0.634
Filling2	0.097	0.088	0.272
Filling3	0.468	0.081	<0.001
Butter1	0.063	0.089	0.478
Filling2:Butter1	0.248	0.119	0.038
Filling3:Butter1	0.277	0.111	0.012

- According to the Wald tests, `Bread2`, `Filling3` and both **interaction** terms are significant at the 5% level.
- But can we trust these standard errors and the associated tests?

Is there overdispersion?

- Relative to the Poisson model, **overdispersion** means that $\mathrm{var}(Y_i) = \phi\,\mu_i$ with $\phi > 1$. Hence, we need an **estimate** of ϕ.
- Recall from Unit B the **moment estimator** of ϕ which, in the Poisson case, is

$$\hat{\phi} = \frac{X^2}{n-p}, \qquad X^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}.$$

  If the Poisson model is correct, we should find $\hat{\phi} \approx 1$.
- In the `Ants` data we obtain $X^2 = 163.07$ on $n - p = 39$ degrees of freedom, hence $\hat{\phi} = 4.18$ (p-value of the X^2 test <0.001).
- The variance is about **four times** the Poisson one, so the Poisson standard errors are **too small**.

> A large X^2 may also be due to a **misspecified mean**, e.g. omitted covariates or interactions, or a wrong link.

A word of caution

- A tempting shortcut is to compare the sample mean and variance of the response. In the `Ants` data they are 43.6 and 327. Is this enough to conclude that there is overdispersion? **No**.
- The relation $\mathrm{var}(Y_i) = \phi\,\mu_i$ concerns Y_i **given the covariates**, since μ_i depends on $\boldsymbol{x}_i$.
- Consider $n = 2000$ observations from

$$Y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i), \qquad \log \mu_i = 1 + 3x_i, \qquad x_i \overset{\text{iid}}{\sim} \text{Bernoulli}(0.5),$$

  which have **no overdispersion** by construction.
- The simulated data have mean 28.2 and variance 691.2, simply because the covariate shifts the mean.
- Once the covariate is accounted for, the Poisson fit gives $X^2/(n-p) = 1.009$.

Quasi-likelihood assumptions

The **quasi-likelihood** approach only relies on the following **second-order** assumptions:

- **(E.1)** $\mathbb{E}(Y_i) = \mu_i = g^{-1}(\boldsymbol{x}_i^T \beta)$;
- **(E.2)** $\text{var}(Y_i) = \dfrac{\phi}{\omega_i} v(\mu_i)$, with $\phi > 0$ **unknown**;
- **(E.3)** Y_i and Y_j are independent for $i \neq j$.

- These assumptions define a **semiparametric** model: only the first two moments of Y_i are specified, the distribution is not.
- A GLM with $Y_i \overset{\text{ind}}{\sim} \text{ED}(\mu_i, a_i(\phi)v(\mu_i))$ and $a_i(\phi) = \phi/\omega_i$ satisfies **(E.1)**-**(E.3)**, but it also specifies the **whole distribution** of Y_i.

On the choice of the variance function

- With identity link and $v(\mu_i) = \omega_i = 1$, **(E.1)-(E.3)** are essentially the Gauss-Markov assumptions of Unit A, with $\phi = \sigma^2$. The novelty concerns **counts** and **proportions**, where GLMs fix $\phi = 1$.
- Keeping the variance function of a GLM, **(E.2)** **inflates** its variance by ϕ: **overdispersion** if $\phi > 1$, **underdispersion** (rare) if $\phi < 1$.
- The two most common quasi-likelihood models are:
  - the **quasi-Poisson** model, with $\mathrm{var}(Y_i) = \phi\,\mu_i$;
  - the **quasi-binomial** model, with $\mathrm{var}(Y_i) = \phi\,\mu_i(1-\mu_i)/m_i$.

As noted in Unit C, the quasi-binomial model makes sense only for **grouped** data. With binary data ($m_i = 1$) we necessarily have $\mathrm{var}(Y_i) = \mu_i(1-\mu_i)$, hence $\phi = 1$.

The variance function $v(\cdot)$ need not be the one of a GLM: e.g. $v(\mu) = \mu^2(1-\mu)^2$ for proportions (Wedderburn 1974) is not the variance function of any **exponential dispersion family**.

The quasi-score

- Without a likelihood, how do we estimate β?
- Recall from Unit B that the **likelihood equations** of a GLM are $q_r(\beta) = 0$, for $r = 1, \ldots, p$, where

$$q_r(\beta) = \sum_{i=1}^{n} \frac{(y_i - \mu_i)}{\mathrm{var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_r} = \frac{1}{\phi} \sum_{i=1}^{n} \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)}.$$

- These equations depend on the distribution of Y_i **only** through $\mathbb{E}(Y_i)$ and $\mathrm{var}(Y_i)$, that is, only through assumptions **(E.1)** and **(E.2)**.
- Hence, we define $\hat{\beta}$ as the solution of $q_r(\beta) = 0$, and we call $q_r(\beta)$ the **quasi-score**. These equations do not depend on ϕ and are solved with the **IRLS** algorithm.
- If $v(\cdot)$ is the variance function of a GLM, as in the quasi-Poisson and quasi-binomial models, $\hat{\beta}$ is (also) the **maximum likelihood estimate** of that GLM.

Properties of the quasi-score

- Let us regard the quasi-score $q_r(\beta)$ as a random variable, i.e. with Y_i in place of y_i.
- Under **(E.1)-(E.3)** alone, it satisfies the **Bartlett identities** of a genuine score function, namely $\mathbb{E}\{q_r(\beta)\} = 0$ and

$$\mathbb{E}\{q_r(\beta)q_s(\beta)\} = -\mathbb{E}\left\{\frac{\partial q_r(\beta)}{\partial \beta_s}\right\} = \sum_{i=1}^{n} \frac{1}{\mathrm{var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_r} \frac{\partial \mu_i}{\partial \beta_s} = [\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X}]_{rs},$$

where $\boldsymbol{W} = \mathrm{diag}(w_1, \ldots, w_n)$ contains the weights of Unit B, that is

$$w_i = \frac{1}{\mathrm{var}(Y_i) g'(\mu_i)^2} = \frac{1}{\phi} \frac{\omega_i}{v(\mu_i) g'(\mu_i)^2}.$$

The quasi-score behaves like a genuine score, even though no distribution has been specified. This explains the name **quasi-likelihood**, and it is the key to the asymptotic results.

Asymptotic properties of $\hat{\beta}$

Under the assumptions **(E.1)**-**(E.3)** and (mild) regularity conditions, the estimator $\hat{\beta}$ is **consistent** and **asymptotically normal**:

$$\hat{\beta} \dot{\sim} \mathrm{N}_p \left(\beta, (\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X})^{-1} \right).$$

- The result is **formally identical** to the one of Unit B, but it holds under much **weaker assumptions**, as for the OLS estimator under the Gauss-Markov assumptions.
- Does it make sense to talk about the **exact** distribution of $\hat{\beta}$? **No**: it would require the distribution of Y_i, which we deliberately left unspecified.
- Consistency only requires **(E.1)**: if the **mean** is correctly specified, $\hat{\beta}$ is consistent even when $v(\mu_i)$ is wrong. The **standard errors**, however, rely on **(E.2)**.

A recipe for overdispersion using quasi-likelihoods

- The dispersion ϕ is estimated with the **method of moments** estimator already used for the `Ants` data, which is **consistent** under **(E.1)**-**(E.3)**. In general:

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{1}{n-p}\sum_{i=1}^{n} \omega_i \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}.$$

- The standard errors are obtained from $(\boldsymbol{X}^T \hat{\boldsymbol{W}} \boldsymbol{X})^{-1}$, plugging in $\hat{\beta}$ and $\hat{\phi}$. The weights w_i are **inversely proportional** to ϕ.

Variance inflation in a nutshell

When $v(\cdot)$ is the variance function of a GLM, as in the quasi-Poisson and quasi-binomial models:

1. Fit the **ordinary GLM** and keep its maximum likelihood estimate $\hat{\beta}$.
2. Multiply the **standard errors** of the GLM by $\sqrt{\hat{\phi}}$, that is $[\widehat{\text{se}}_{\text{quasi}}(\hat{\beta})]_j = \sqrt{\hat{\phi}}\,[\widehat{\text{se}}_{\text{glm}}(\hat{\beta})]_j$.

The quasi-log-likelihood

- To compare nested models we need something like a log-likelihood. Under **(E.1)**-**(E.3)** we define the **quasi-log-likelihood**

$$\ell_Q(\beta, \phi) = \sum_{i=1}^{n} \int_{y_i}^{\mu_i} \frac{\omega_i (y_i - t)}{\phi\, v(t)}\, \mathrm{d}t, \qquad \mu_i = g^{-1}(\boldsymbol{x}_i^T \beta),$$

  provided that these integrals exist.
- Its gradient is the **quasi-score**, that is $\partial \ell_Q(\beta, \phi)/\partial \beta_r = q_r(\beta)$. Indeed, by the chain rule, the derivative of the ith integral is $\omega_i (y_i - \mu_i)/\{\phi\, v(\mu_i)\} \times \partial \mu_i / \partial \beta_r$.
- Since $\ell_Q(\beta, \phi)$ is a function of β, its second derivatives $\partial q_r(\beta)/\partial \beta_s$ are **symmetric** in r and s, as for a genuine log-likelihood.
- In general $\ell_Q(\beta, \phi)$ is **not** a log-likelihood. However, in the Poisson case ($\omega_i = 1$, $v(t) = t$, $\phi = 1$) we get $\ell_Q(\beta, 1) = \sum_{i=1}^{n} (y_i \log \mu_i - \mu_i)$, up to constants, which is the **Poisson log-likelihood**.

Comparing nested models

- The quantity $D(\boldsymbol{y};\boldsymbol{\mu}) = -2\phi\,\ell_Q(\beta,\phi)$ is called **quasi-deviance**. When $v(\cdot)$ is the variance function of a GLM, it coincides with the **deviance** of Unit B, e.g. the Poisson or the binomial one.
- Let $M_0 \subset M_1$ be two **nested models**, with p_0 and p parameters, and let $q = p - p_0$. The **quasi-log-likelihood ratio** statistic is

$$W = 2[\ell_Q(\hat\beta,\hat\phi) - \ell_Q(\hat\beta_0,\hat\phi)] = \frac{D(\boldsymbol{y};\hat{\boldsymbol{\mu}}_0) - D(\boldsymbol{y};\hat{\boldsymbol{\mu}})}{\hat\phi} \,\dot\sim\, \chi^2_q,$$

  where $\hat\phi$ is estimated from the **larger** model.

- The (quasi-)deviances must **not** be used as goodness-of-fit statistics, since ϕ is unknown.
- There is no likelihood, therefore the **AIC** is **not available**.

The quasi-Poisson model

- With the same mean structure, we obtain $\hat{\phi} = 4.18$: the standard errors are multiplied by $\sqrt{\hat{\phi}} = 2.04$.

	Estimate	SE (Poisson)	SE (quasi)	p-value (Poisson)	p-value (quasi)
(Intercept)	3.386	0.075	0.153	<0.001	<0.001
Bread2	0.154	0.061	0.124	0.011	0.221
Bread3	0.026	0.063	0.128	0.684	0.843
Bread4	-0.030	0.064	0.130	0.634	0.817
Filling2	0.097	0.088	0.180	0.272	0.594
Filling3	0.468	0.081	0.166	<0.001	0.008
Butter1	0.063	0.089	0.182	0.478	0.730
Filling2:Butter1	0.248	0.119	0.244	0.038	0.316
Filling3:Butter1	0.277	0.111	0.226	0.012	0.229

- The **estimates** are identical, but the standard errors are larger: the tests are affected.

Model simplification

- The larger standard errors suggest a **simpler** model. We remove one term at a time, using the quasi-log-likelihood ratio statistic W and its χ^2_q approximation.

Model	Removed term	q	W	p-value
Full model	Bread	3	2.581	0.461
Full model	Filling:Butter	2	1.621	0.445
Without Bread	Filling:Butter	2	1.614	0.446

- Neither `Bread` nor the `Filling`-`Butter` interaction is relevant, once overdispersion is accounted for.
- The final model only includes `Filling` and `Butter`, with $p = 4$ parameters.

The final model

	Estimate	Std. Error	p-value	exp(Estimate)
(Intercept)	3.318	0.103	$<$0.001	27.596
Filling2	0.233	0.120	0.059	1.262
Filling3	0.621	0.111	$<$0.001	1.860
Butter1	0.263	0.089	0.005	1.301

- Under the logarithmic link, $\exp(\beta_j)$ is the **multiplicative change** of the mean number of ants (Unit D). The estimated dispersion is $\hat{\phi} = 4.07$.
- Compared to Vegemite, **ham and pickles** attract on average about 86% more ants, and peanut butter about 26% more (borderline significant).
- The presence of **butter** increases the mean number of ants by about 30%. The type of bread is irrelevant.

References

Agresti, A. (2015), *Foundations of Linear and Generalized Linear Models*, Wiley.

Salvan, A., Sartori, N., and Pace, L. (2020), *Modelli lineari generalizzati*, Springer.

Wedderburn, R. W. M. (1974), "Quasi-likelihood functions, generalized linear models, and the gauss-newton method," *Biometrika*, 61, 439–447.